Student Answer Script Transcription (High-Resolution Stitched Document)

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Q-1) Space for answering Question-1 only.

a) For S to be subspace of $M_{4\times4}(R)$, it must satisfy the following:

i) S must include zero matrix.

In zero matrix, as all the elements are zero, the condition for S is satisfied (0+0+0 = 0) for fourth column.

Closure Under Addition

Let W_1 and W_2 be two elements from S. Then $W_1 + W_2$ must belong to S.

Each element in $W_1 + W_2$ is the sum of respective elements of W_1 and W_2. So, the fourth column of $(W_1 + W_2)$ = Fourth col. of W_1 + Fourth col. of W_2, which can be further broken down to sum of first 3 columns of W_1 + sum of first 3 columns of W_2. By associativity of addition, $(W_1 + W_2)$ belongs to S.

Closure Under Scalar Multiplication

If $W \in S$, then tW must belong to S. As in row $\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{bmatrix}$, $ta_1 + ta_2 + ta_3 = t(a_1 + a_2 + a_3)$ (distributive and commutative).